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CHE F301 Kinetics & Reactor Design

Algorithm

Heat Effects

Isothermal Design

Stoichiometry

Rate Laws

Mole Balance

CRE Algorithm

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Conversion

$aA + bB \rightarrow cC + dD$

Taking A as a basis, $A + \frac{b}{a}B \rightarrow \frac{c}{a}C + \frac{d}{a}D$

Conversion, X_A , is the number of moles A that have reacted per mole of A fed to the system

$$X_A = \frac{\text{Moles of A reacted}}{\text{Moles of A fed}}$$

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Batch Reactor

$$\left[\begin{matrix} \text{Moles A} \\ \text{remaining} \end{matrix} \right] = \left[\begin{matrix} \text{Moles A} \\ \text{Initially} \end{matrix} \right] - \left[\begin{matrix} \text{Moles A} \\ \text{reacted} \end{matrix} \right]$$
$$N_A = N_{A0} - N_{A0}X$$

$$dN_A = 0 - N_{A0}dX$$
$$\frac{dN_A}{dt} = -N_{A0} \frac{dX}{dt} = r_A V$$

Batch time, t , required to achieve a conversion X .



Integrating,

$$t = N_{A0} \int_0^X \frac{dX}{-r_A V}$$

The necessary t to achieve conversion X .

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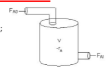
CH 1511 Kinetics & Reactor Design

CSTR Reactor

For a rxn: $A \xrightarrow{k} B \rightarrow C \xrightarrow{k} D$

Recall Design Equation for CSTR (Chapter 1);

$$V = \frac{F_{A0} - F_A}{-r_A} \quad [1]$$



Substituting $F_A = F_{A0} - F_{A0}X$ into [1]

$$V = \frac{F_{A0} - (F_{A0} - F_{A0}X)}{-r_A}$$

Rearranging;

$$V = \frac{F_{A0}X}{-r_A}$$

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Plug Flow Reactor

Recall Mole Balance for PFR

$$\frac{dF_A}{dV} = -r_A \quad [1]$$



We know that

$$F_A = F_{A0} - F_{A0}X \quad [2]$$

Differentiating [2] wrt X

$$\frac{dF_A}{dX} = 0 - F_{A0} \Rightarrow dF_A = -F_{A0}dX \quad [3]$$

Substituting [3] into [1]

$$F_{A0} \frac{dX}{dV} = -r_A \quad [4]$$

$$V = F_{A0} \int_0^X \frac{dX}{-r_A}$$

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Packedbed Reactor

Design equation for PBR;

$$F_{A0} \frac{dX}{dW} = -r_A$$



Similar to that of PFR except these terms:

Catalyst weight ;

$$W = F_{A0} \int_0^X \frac{dX}{-r_A}$$

$$V \rightarrow W$$
$$-r_A \rightarrow -r_A$$

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Summary of Reactor Mole Balance

Reactor	Differential Form	Algebraic Form	Integral Form
Batch	$N_{A0} \frac{dX}{dt} = -r_A V$		$t = N_{A0} \int_0^X \frac{dX}{-r_A V}$
CSTR	-	$V = \frac{F_{A0} X}{-r_A}$	-
PFR	$F_{A0} \frac{dX}{dV} = -r_A$		$V = F_{A0} \int_0^X \frac{dX}{-r_A}$
PBR	$F_{A0} \frac{dX}{dW} = -r_A'$		$W = F_{A0} \int_0^X \frac{dX}{-r_A'}$

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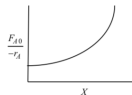
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Levenspiel Plots

➤ Reactor sizing

- CSTR/ PFR – for a given conversion- Volume?
- Plot of $(F_{A0}/-r_A)$ or $(1/-r_A)$ as a function of X

$$\frac{F_{A0}}{-r_A} = g(X)$$



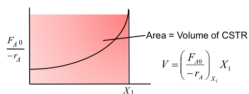
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CSTR



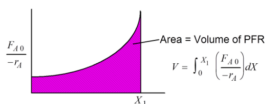
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PFR



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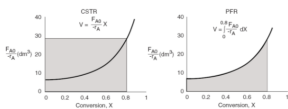
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CSTR/PFR Sizing

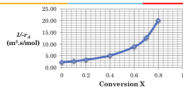


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Use plot of $1/r_A$ vs X to size flow reactors for different entering molar flow rates, F_{A0}

Important Notes (For Irreversible Rxn, $A \rightarrow B+C$):

1. If the reaction is carried out isothermally, the rate is usually greatest at the start of the reaction, when the concentration is greatest (when $X=0$, $1/r_A$ is small (r_A is big)).
2. As $X \rightarrow 1$, $-r_A \rightarrow 0$, thus $1/r_A \rightarrow \infty$, $V \rightarrow \infty$

An infinite reactor volume is needed to reach complete conversion

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Reactor Sizing

Important Notes (cont):

(For Reversible Rxn, $A \rightleftharpoons B+C$):

1. The **max conversion** is the equilibrium conversion, X_e .
2. At equilibrium, $r_d(\text{net}) = 0$.

 $X \rightarrow X_e$, $-r_A \rightarrow 0$, thus $1/r_A \rightarrow \infty$, $V \rightarrow \infty$

An infinite reactor volume is needed to reach X_e

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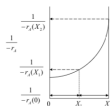
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Numerical Evaluations of Integrals

- Simpson's One-Third Rule:



$$V = \int_0^{X_e} \frac{F_{A0}}{-r_A} dX \approx \frac{\Delta X}{3} F_{A0} \left[\frac{1}{-r_A(0)} + \frac{4}{-r_A(X/2)} + \frac{1}{-r_A(X_e)} \right]$$

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Reactor Sizing: Sizing A PFR

Trapezoidal Rule (2-point):

$$\int_{X_0}^{X_1} f(X) dX = \frac{h}{2} [f(X_0) + f(X_1)]$$

$$\text{where } h = X_1 - X_0$$

Simpson's Three-Eighths Rule (4-point):

$$\int_{X_0}^{X_3} f(X) dX = \frac{3h}{8} [f(X_0) + 3f(X_1) + 3f(X_2) + f(X_3)]$$

$$\text{where } h = \frac{X_3 - X_0}{3} \quad X_1 = X_0 + h \quad X_2 = X_0 + 2h$$

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Simpson's Five-Point Quadrature Formula:

$$\int_{X_0}^{X_4} f(X) dX = \frac{h}{3} [f(X_0) + 4f(X_1) + 2f(X_2) + 4f(X_3) + f(X_4)]$$

$$\text{where } h = \frac{X_4 - X_0}{4}$$

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CSTR/ PFR sizing

Problem:

Let's consider the isothermal gas-phase isomerization:
A → B

X	$-r_A$ (mol/m ³ s)
0	0.45
0.1	0.37
0.2	0.30
0.4	0.195
0.6	0.113
0.7	0.079
0.8	0.05

$$[T = 500 \text{ K}]$$

$$[P = 820 \text{ kPa} = 6.2 \text{ atm}]$$

initial charge was pure A

$$V_{\text{CSTR}} = \frac{F_{A0} \cdot X}{-r_A}$$

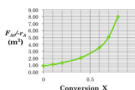
Calculate the volume required to achieve 80% conversion in a CSTR & PFR. Given, species A enters the reactor at a molar flow rate of 0.4 mol/s.

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X	0	0.1	0.2	0.4	0.6	0.7	0.8
r_A	0.45	0.37	0.3	0.195	0.113	0.079	0.05
F_{A0}/r_A	0.89	1.08	1.33	2.05	3.54	5.06	8.00



$$\left(-\frac{1}{r_A} \right)_{X=0.8} = \frac{1}{0.05 \text{ mol/m}^3 \cdot \text{s}} = 20 \frac{\text{m}^3 \cdot \text{s}}{\text{mol}}$$

2. Calculate V .

$$V = \frac{F_{A0} \cdot X}{-r_A} = 0.4 \frac{\text{mol}}{\text{s}} \left(\frac{20 \text{ m}^3 \cdot \text{s}}{\text{mol}} \right) (0.8) = 6.4 \text{ m}^3$$

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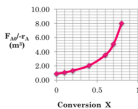
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For PFR

X	0	0.1	0.2	0.4	0.6	0.7	0.8
r_A	0.45	0.37	0.3	0.195	0.113	0.079	0.05
F_{A0}/r_A	0.89	1.08	1.33	2.05	3.54	5.06	8.00



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Recall 5-Point Quadrature Rule:

$$\int_{X_0}^{X_4} f(X) dX = \frac{h}{3} [f(X_0) + 4f(X_1) + 2f(X_2) + 4f(X_3) + f(X_4)]$$

$$\text{where } h = \frac{X_4 - X_0}{4}$$

Find h (ΔX):

$$h = \frac{0.8 - 0}{4} = 0.2$$

$$X_0 = 0, X_1 = 0.2, X_2 = 0.4, X_3 = 0.6, X_4 = 0.8$$

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Find V:

$$V = \frac{\Delta X}{3} \left[\frac{F_{R0}}{-r_A(X=0)} + 4 \frac{F_{R0}}{-r_A(X=0.2)} + 2 \frac{F_{R0}}{-r_A(X=0.4)} + 4 \frac{F_{R0}}{-r_A(X=0.6)} + \frac{F_{R0}}{-r_A(X=0.8)} \right]$$

Substituting the numerical values:

$$V = \frac{0.2}{3} [0.89 + 4(1.33) + 2(2.05) + 4(3.54) + 8.0] \text{ m}^3$$

$$= 2.165 \text{ m}^3$$

→ PFR with volume of 2.165 m³ is required to reach 80% conversion

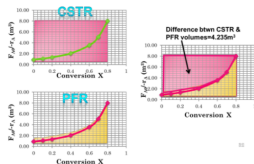
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COMPARING VOLUME OF CSTR & PFR



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$V_{CSTR} > V_{PFR}$ for the same conversion & rxn condition.

WHY???

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Reactors in series

- Given: r_A as a function of **conversion**

$$X_i = \frac{\text{Total moles of A reacted up to point } i}{\text{Moles of A fed to the first reactor}}$$

Molar Flow rate of species A at point i:

$$F_{Ai} = F_{A0} - F_{A0} X_i$$

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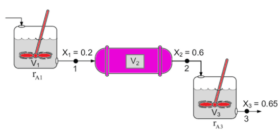
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Reactors in series



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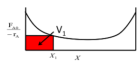
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Reactor 1:

$$F_{A1} = F_{A0} - F_{A0} X_1$$

$$V_1 = \frac{F_{A0} - F_{A1}}{-r_{A1}} = \frac{F_{A0} - (F_{A0} - F_{A0} X_1)}{-r_{A1}} = \frac{F_{A0} X_1}{-r_{A1}}$$



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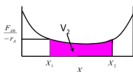
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Reactor 2:

$$V_2 = \int_{X_1}^{X_2} \frac{F_{A0} dX}{-r_A}$$



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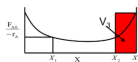


Reactor 3:

$$F_{A2} - F_{A3} + r_{A3}V_3 = 0$$

$$(F_{A0} - F_{A2}X_2) - (F_{A0} - F_{A3}X_3) + r_{A3}V_3 = 0$$

$$V_3 = \frac{F_{A0}(X_3 - X_2)}{-r_{A3}}$$

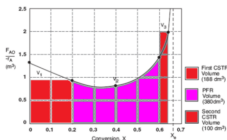


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CSTRs in series

Reactor 1:

Mole Balance:

In - Out + Generation = 0

$$F_{A0} - F_{A1} + r_{A1}V_1 = 0 \quad [1]$$

The molar flow rate of A at point 1:

$$F_{A1} = F_{A0} - F_{A0}X_1 \quad [2]$$

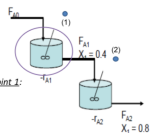
Combining [1] & [2]:

$$V_1 = F_{A0} \left(\frac{1}{-r_{A1}} \right) X_1$$

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**Reactor 2:**Mole Balance:

In – Out + Generation = 0

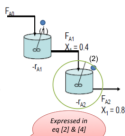
$$F_{A1} - F_{A2} + r_{A2}V_2 = 0 \quad [3]$$

The molar flow rate of A at point 2:

$$F_{A2} = F_{A0} - F_{A0}X_2 \quad [4]$$

Combining [3] & [4]:

$$V_2 = \frac{F_{A0} - F_{A0}X_2}{-r_{A2}} = \frac{F_{A0}(1 - X_2)}{-r_{A2}} \quad [5]$$



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$$F_{A1} = F_{A0} - F_{A0}X_1 \quad [2]$$

$$F_{A2} = F_{A0} - F_{A0}X_2 \quad [4]$$

$$V_2 = \frac{F_{A1} - F_{A2}}{-r_{A2}} \quad [5]$$

Substituting [2] & [4] into [5]:

$$\begin{aligned} V_2 &= \frac{F_{A0} - F_{A0}X_1 - (F_{A0} - F_{A0}X_2)}{-r_{A2}} \\ &= \frac{-F_{A0}X_1 + F_{A0}X_2}{-r_{A2}} = \frac{F_{A0}}{-r_{A2}} (X_2 - X_1) \end{aligned}$$

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**Problem**

X	0.0	0.1	0.2	0.4	0.6	0.7	0.8
$(F_{A0}/r_{Ai})[\text{m}^3]$	0.89	1.08	1.33	2.05	3.54	5.06	8.0

For the two CSTRs in series, 40% conversion is achieved in the first reactor. What is the volume of each of the two reactors necessary to achieve 80% overall conversion of entering species?



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**Problem**For reactor 1, $X = 0.4$

$$\begin{aligned} V_1 &= F_{A0} \left(\frac{1}{-r_{A1}} \right) X_1 \\ &= \left(\frac{F_{A0}}{-r_{A1}} \right)_{X=0.4} \cdot X_1 = 2.05 \text{m}^3 (0.4) = 0.82 \text{m}^3 \end{aligned}$$

For reactor 2, $X = 0.8$

$$\begin{aligned} V_2 &= \left(\frac{F_{A0}}{-r_{A2}} \right)_{X=0.8} (X_2 - X_1) \\ &= (8.0 \text{m}^3) (0.8 - 0.4) = 3.2 \text{m}^3 \end{aligned}$$

$$\begin{aligned} \text{Total } V &= (0.82 + 3.2) \text{m}^3 \\ &= 4.02 \text{m}^3 \end{aligned}$$



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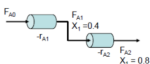
CHE 3331 Kinetics & Reactor Design

Pfr in Series

Problem

X	0.0	0.1	0.2	0.4	0.6	0.7	0.8
$[F_{A0}/r_A](\text{m}^3)$	0.89	1.08	1.33	2.05	3.54	5.06	8.0

Calculate the reactor volume V_1 and V_2 for the plug-flow sequence shown below when the intermediate conversion is 40% & the final conversion is 80%.



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X	0.0	0.1	0.2	0.4	0.6	0.7	0.8
$[F_{A0}/r_A](\text{m}^3)$	0.89	1.08	1.33	2.05	3.54	5.06	8.0

$$\int_0^X f(X) dX = \frac{b}{3} [f(X_2) + 4f(X_1) + f(X_0)]$$

$$\text{where } h = \frac{X_2 - X_1}{2}, X_1 = X_0 + h$$

For reactor 1, $\Delta X = 0.2$, $X_0 = 0$, $X_1 = 0.2$, $X_2 = 0.4$

$$V_1 = \int_0^{0.4} \frac{F_{A0}}{-r_A} dX = \frac{\Delta X}{3} \left[\frac{F_{A0}}{-r_A(0)} + 4 \frac{F_{A0}}{-r_A(0.2)} + \frac{F_{A0}}{-r_A(0.4)} \right]$$

$$= \left(\frac{0.2}{3} \right) [0.89 + 4(1.33) + 2.05] \text{ m}^3 = 0.55 \text{ m}^3$$

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For reactor 2, $\Delta X = 0.2$, $X_0 = 0.4$, $X_1 = 0.6$, $X_2 = 0.8$

$$V_2 = \int_{0.4}^{0.8} \frac{F_{A0}}{-r_A} dX = \frac{\Delta X}{3} \left[\frac{F_{A0}}{-r_A(0.4)} + 4 \frac{F_{A0}}{-r_A(0.6)} + \frac{F_{A0}}{-r_A(0.8)} \right]$$

$$= \left(\frac{0.2}{3} \right) [2.05 + 4(3.54) + 8.0] \text{ m}^3 = 1.614 \text{ m}^3$$

Total volume;

$$V = V_1 + V_2 = 0.551 + 1.614 \text{ m}^3 = 2.165 \text{ m}^3$$

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CHE 3331 Kinetics & Reactor Design

Flow reactors performance

- **Space time:** Time required to process one reactor volume of feed measured at specified conditions

- A space-time of 2 min means that every 2 min one reactor volume of feed at specified conditions is being treated by the reactor.

- **Space velocity:** number of reactor volumes of feed at specified conditions which can be treated in unit time

- A space-velocity of 5 hr^{-1} means that five reactor volumes of feed at specified conditions are being fed into the reactor per hour.

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CHE 5521 Kinetics & Reactor Design



$$\tau = \frac{1}{s} = \frac{C_{A0}V}{F_{A0}} = \frac{\left(\frac{\text{moles A entering}}{\text{volume of feed}}\right) (\text{volume of reactor})}{\left(\frac{\text{moles A entering}}{\text{time}}\right)}$$

$$= \frac{V}{v_0} = \frac{(\text{reactor volume})}{(\text{volumetric feed rate})}$$

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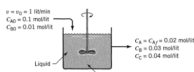
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CHE 5521 Kinetics & Reactor Design



- One liter per minute of liquid containing A and B ($C_{A0} = 0.10$ mol/liter, $C_{B0} = 0.01$ mol/liter) flow into a mixed reactor of volume $V = 1$ liter. The materials react in a complex manner for which the stoichiometry is unknown. The outlet stream from the reactor contains A, B, and C ($C_A = 0.02$ mol/liter, $C_B = 0.03$ mol/liter, $C_C = 0.04$ mol/liter), as shown in Fig. Find the rate of reaction of A, B, and C for the conditions within the reactor.

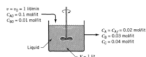


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CHE 5521 Kinetics & Reactor Design



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PFRs in Series and/or in Parallel

N PFRs are connected in series, the balance for the P^{th} reactor

$$\frac{V}{F_0} = \int_{X_{P-1}}^{X_P} \frac{dX}{-r}$$

For N reactors in series

$$\frac{V}{F_0} = \sum_{P=1}^N \frac{V_P}{F_0} = \frac{V_1 + V_2 + \dots + V_N}{F_0}$$

$$= \int_{X_0=0}^{X_1} \frac{dX}{-r} + \int_{X_1}^{X_2} \frac{dX}{-r} + \dots + \int_{X_{N-1}}^{X_N} \frac{dX}{-r} = \int_0^{X_N} \frac{dX}{-r}$$

Hence, N plug flow reactors in series with a total volume V gives the same conversion as a single plug flow reactor of volume V .

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- Plug flow reactors connected in parallel or in any parallel-series combination, we can treat the whole system as a single plug flow reactor of volume equal to the total volume of the individual units if the feed is distributed in such a manner that fluid streams that meet have the same composition. Thus, for reactors in parallel V/F or τ must be the same for each parallel line. Any other way of feeding is less efficient

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PFR-series/parallel

- The reactor setup shown in Fig. consists of three plug flow reactors in two parallel branches. Branch D has a reactor of volume 50 liters followed by a reactor of volume 30 liters. Branch E has a reactor of volume 40 liters. What fraction of the feed should go to branch D?



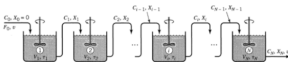
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Equal-Size MFRs in series



For first order reaction, constant density system

$$\tau_1 = \frac{C_{i0} V_1}{F_{i0}} = \frac{V}{D} = \frac{C_{i0}(X_1 - X_{i-1})}{-r_{A1}}$$

$$\tau_1 = \frac{C_{i0}((1 - C/C_0) - (1 - C_{i-1}/C_0))}{k C_1} = \frac{C_{i-1} - C_1}{k C_1} \Rightarrow \frac{C_{i-1}}{C_1} = 1 + k \tau_1$$

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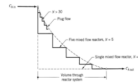
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$$\frac{C_1}{C_0} = \frac{1}{1 + X_1} = \frac{C_0 C_1}{C_1 C_2} \dots \frac{C_{N-1}}{C_N} = (1 + k \tau_1)^N$$

$$\tau_{\text{reaction}} = N \tau_1 = \frac{N}{k} \left[\left(\frac{C_0}{C_N} \right)^{1/N} - 1 \right]$$

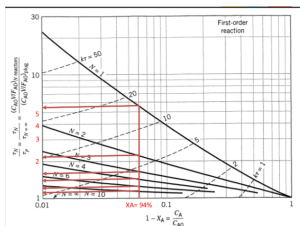
If $N \rightarrow \infty$ $\tau_1 = \frac{1}{k} \ln \frac{C_0}{C_N}$ this equation is same as 1st order PFR eqN



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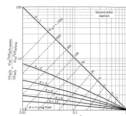
Equal size MFRs-2nd Order Reactions

N reactors in series

$$C_N = \frac{1}{48\eta} \left(-2 + 2\sqrt{-1 + \dots + 2\sqrt{-1 + 2\sqrt{1 + 4C_0 k \tau}} N} \right)$$

N → on Plug flow

$$\frac{C_0}{C} = 1 + C_0 k \tau$$



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MFRs of different size

➤ For arbitrary kinetics in mixed flow reactors of different size, two types of questions may be asked:

- How to find the outlet conversion from a given reactor system?
- How to find the best setup to achieve a given conversion?

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MFRs Different Sizes



$$\eta_1 = \bar{t}_1 = \frac{V_1}{F_1} = \frac{C_0 - C_1}{(-r)_1} \Rightarrow \frac{1}{\eta_1} = \frac{(-r)_1}{C_0 - C_1}$$

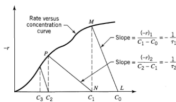
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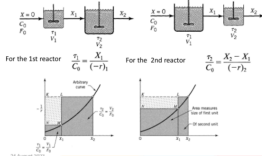
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Best arrangement?

To find the minimum size of two mixed flow reactors for a specified conversion



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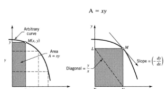
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Maximization of Rectangles



- This area is maximized when

$$dA = 0 = y \, dx + x \, dy \rightarrow -dy/dx = y/x$$

- The area is maximized when M is at that point where the slope of the curve equals the slope of the diagonal NL of the rectangle. Valid mostly for $n > 0$

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Optimum case

The optimum size ratio of the two reactors is achieved where the slope of the rate curve at M equals the diagonal NL.



For the special case of first-order reactions equal-size reactors are best; for reaction orders $n > 1$ the smaller reactor should come first; for $n < 1$ the larger should come first

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Best arrangement

1. If $1/-r_A$ vs C_A curve rises monotonically (order, $n > 0$) the reactors should be connected in series. They should be ordered so as to keep the concentration of reactant as high as possible if the rate-concentration curve is convex ($n > 1$), and as low as possible if the curve is concave ($n < 1$).

2. The order should be **plug, small mixed, large mixed**, for $n > 1$; the reverse order should be used when $n < 1$.

3. For reactions where the rate-concentration curve passes through a maximum or minimum the arrangement of units depends on the actual shape of curve, the conversion level desired, and the units available. **No simple rules can be suggested.**

4. The best way is: $1/-r_A$ vs. C_A curve

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P2-7. The exothermic reaction $A \rightarrow B + C$ was carried out adiabatically and the following data recorded. The entering molar flowrate of A was 300 mol/min.

(a) What are the PFR and CSTR volumes necessary to achieve 40% conversion?

(b) Over what range of conversions would the CSTR and PFR reactor volumes be identical?

(c) What is the maximum conversion that can be achieved in a 105-dm³ CSTR?

(d) What conversion can be achieved if a 72-dm³ PFR is followed in series by a 24-dm³ CSTR?

(e) What conversion can be achieved if a 24 dm³ CSTR is followed in a series by a 72-dm³ PFR?

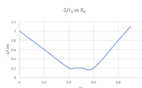
X	0	0.2	0.4	0.45	0.5	0.6	0.8	0.9
$-r_A$ (mol/dm ³ min)	1.0	1.67	5.0	5.0	5.0	5.0	1.25	0.91

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